

Mohammad Alkhalifah

✉ mohammad.alkhalifah@icloud.com in MKHLF 🔗 MKHLF

Education

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| King Abdullah University of Science and Technology
<i>M.S. Computer Science — CGPA: 3.8</i>
<ul style="list-style-type: none"> ◦ Recipient of King Abdullah University of Science and Technology Gifted Student Program Scholarship (KGSP) ◦ Coursework: GPU and GPGPU, Computing Systems and Concurrency | Thuwal, SA
<i>Present</i> |
| University of California San Diego
<i>B.S. Computer Science — CGPA: 3.9</i>
<ul style="list-style-type: none"> ◦ Coursework: Deep Learning, Computer Vision, Compilers Construction, Parallel Computing, Computer Networks, Computer Architecture: Software Perspective, Operating Systems, Ubiquitous Computing | La Jolla, CA
<i>Dec 2024</i> |

Experience

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|---|---|
| AI Engineer
<i>Sawt</i> 🔗 | Riyadh, SA
<i>Jan 2026 – Present</i> |
| <ul style="list-style-type: none"> ◦ Built and deployed a production Conformer-CTC ASR system for Saudi-Arabic telephony, from entity-aware benchmarking and multi-model transcription fusion over 900 hours of speech to training on 1,500+ hours; ranked first on the public Open Universal Arabic ASR Leaderboard and led the Sawt Telephonic Benchmark against leading commercial APIs, achieving 10.5% WER at 49 ms median latency ◦ Adapted the ASR system to a domain-specific production ordering workflow using acoustic fine-tuning, synthetic augmentation, and vocabulary-aware decoding, reducing WER from 25.6% to 16.5% and increasing entity recall from 67.4% to 82.1% ◦ Built an Arabic evaluation and synthetic-data stack for multi-turn, tool-using voice agents, designing a multi-domain benchmark from the ground up and mining patterns from 500,000+ production calls to generate 30,000+ quality-gated training conversations | |
| Machine Learning Scientist
<i>Intelmatix</i> 🔗 | Hybrid
<i>Jan 2025 – Oct 2025</i> |
| <ul style="list-style-type: none"> ◦ Led development of data analysis features for Site Selection product, including temporal delta analysis for detecting market trends and a customizable key-attraction scoring engine for diverse clients' requirements ◦ Developed automated testing suite using deepEval for SiteAdvisor's LLM capabilities, validating location intelligence recommendations and ensuring consistent performance across diverse market analysis scenarios (Python, deepEval) | |
| Research Intern
<i>Saudi Aramco EXPEC Advanced Research Center</i> | Thuwal, SA
<i>Jun 2024 – Aug 2024</i> |
| <ul style="list-style-type: none"> ◦ Adapted MOMENT Time-series foundation model to assess feasibility of seismic foundational models, achieving 0.90 correlation on trace reconstruction using 0.02% of available data, demonstrating potential for rapid model adaptation across seismic tasks with significantly reduced data requirements ◦ Engineered an end-to-end modular pipeline for large-scale seismic data, enabling comprehensive model development and evaluation across multiple subsurface analysis tasks (Python, PyTorch) | |
| Research Intern
<i>ARMOR Lab, UCSD</i> 🔗 | La Jolla, CA
<i>Jun 2023 – Aug 2023</i> |
| <ul style="list-style-type: none"> ◦ Designed a golf swing analysis system using Motion Tape sensors across 4 key body positions, processing golf swings data creating digital athlete twins for personalized feedback ◦ Enhanced system performance from 54% to 88% accuracy by implementing LSTM models and developing a cross-channel analysis pipeline using CPSD and PCA techniques for improved swing classification (Python, PyTorch) | |
| Academic Tutor
<i>CSE Department: Computer Architecture, Mathematics for Algorithms & Systems</i> | La Jolla, CA
<i>Sep 2022 – Dec 2024</i> |

Projects

Living Worlds: GPU-Accelerated Terrain Simulation [↗](#)

- Developed real-time terrain simulation using cellular automata for erosion modeling and 9-type biome dynamics with bidirectional feedback, achieving +200 FPS at +9M vertices through Vulkan compute/graphics pipeline with ping-pong buffer synchronization (Vulkan, C++)

Deep Reinforcement Learning Strategies for 2048 Game AI [↗](#)

- Implemented and evaluated multiple AI approaches (DQL, MCTS, N-Tuple Networks), achieving 98.2% success rate for 2048 tile through systematic strategy optimization and performance analysis (Python, PyTorch)

User Intent Classification with BERT and Contrastive Learning [↗](#)

- Fine-tuned BERT for user intent classification on Amazon dataset, implementing advanced training techniques (SWA, warm-up steps) and contrastive learning methods (SupContrast, SimCLR), achieving 89% accuracy (Python, PyTorch)

Three-Phase Compiler, Tiger Language [↗](#)

- Designed and implemented a complete compiler pipeline including static optimizer for Tiger-IR, MIPS32 code generator with register allocation (achieving 29-38% memory load reduction), and ANTLR-based lexer/parser with transformed LL(1) grammar and error recovery (Java)

Keep-It-Flow [↗](#)

- Built a browser extension for automated background music management with 30+ weekly active users, managing tabs audio for focused study/work sessions (JavaScript)

TetrisBattle Artificial Intelligence [↗](#)

- Designed a Tetris AI with heuristic-based evaluation for optimal piece placement (Python, NumPy)

FPGA Tetris [↗](#)

- Implemented full-featured Tetris on Artix-7 FPGA with Super Rotation System (wall kicks, T-spin detection), 7-bag randomizer, and 15-state FSM across 5 clock domains, featuring 4-stage pipelined renderer for VGA 1280×800 @ 60Hz output with PS/2 keyboard input (SystemVerilog, Vivado)

Extracurricular Activities

Competition Finalist

La Jolla, CA

Element.AI [↗](#)

Feb 2023

- Led team to second place among 40+ teams in a Snake.io variant competition, developing an AI agent using heuristic-based evaluation functions for best move under a strict 5-hour time constraint

Founder & President

La Jolla, CA

Saudi Student Association at UCSD [↗](#)

Dec 2022 – Oct 2023

- Led UCSD's Saudi Student Association (100+ members), building strong community connections through various events from weekly social gatherings to large-scale multi-university cultural celebrations across Southern California

Junior Analyst

La Jolla, CA

Student Foundation Investment Committee [↗](#)

Oct 2021 – Sep 2022

- Analyzed and presented market trends, financial models, and risk analyses for investment opportunities in UCSD's \$1B+ student-managed fund, contributing to portfolio optimization decisions

Skills

Languages & Technologies: Python, C/C++, PyTorch, CUDA, Vulkan, Git, Linux, Unit Testing, Docker